

Responsible AI: Bias, Fairness, Governance and Sustainability Report— Smart City Traffic Intelligence

1. Data Coverage and Sampling Limitations

The traffic dataset does not provide complete and equal coverage across all years. Some years contain only partial-year observations, which means certain months, seasons and traffic conditions may be represented more heavily than others. In addition, the dataset represents traffic on a single corridor rather than an entire transport network.

These limitations may introduce sampling bias because the machine learning models learn only from the traffic conditions available in the dataset. Therefore, the models may not generalise equally well to other roads, locations, seasons or future traffic patterns. Before real-world deployment, the system should be validated using more recent, complete and geographically diverse traffic data.

2. Proxy Label and Associated Risks

A real accident dataset was not provided for this capstone, so a proxy high_risk label was created to demonstrate the classification workflow. A record was classified as high risk when High or Severe congestion occurred together with severe weather or low-visibility conditions.

This proxy does not represent actual accident occurrence and should not be interpreted as a real accident-prediction target. Because the label is constructed from traffic and weather conditions, the classification model may achieve strong performance partly because the input features are closely related to the rules used to create the label. Therefore, high accuracy or ROC AUC does not prove that the model can predict real-world accidents.

Before any real-world safety use, the model should be retrained and validated using genuine accident records, including reliable information on accident occurrence, severity, location and time.

3. Uneven Model Errors Across Conditions

Model errors may not be distributed equally across all traffic conditions. Performance could vary by time of day, weekday versus weekend, weather condition or congestion level. For example, uncommon weather conditions such as fog, squall or heavy snow may have fewer historical observations than common conditions such as clear or cloudy weather.

As a result, the model may perform better for well-represented traffic conditions and less reliably for rare conditions. This creates a fairness and reliability concern because

some journeys or operating conditions may receive less accurate predictions than others.

Before real-world use, model performance should therefore be evaluated separately across important subgroups, such as time periods, weather categories and congestion levels. If large performance differences are found, additional data collection, model retraining or condition-specific safeguards may be required.

4. Governance and Sustainability in Mobility AI

Before this traffic-prediction system is trusted for real-world mobility decisions, appropriate human oversight should be maintained. Traffic planners or transport operations staff should review model performance, limitations and monitoring alerts before using its predictions to influence traffic-management decisions or traveller recommendations. The system should be treated as decision support rather than an automatic authority. Particular attention should be given to unusual conditions such as severe weather, road incidents or traffic patterns that differ from the historical training data. Model updates should also be tested and approved before deployment, with clear responsibility for responding to monitoring alerts and inaccurate predictions.

Sustainability should also be considered when selecting modelling approaches. In this project, the Random Forest Regressor achieved slightly better performance than the neural network while requiring a simpler training and deployment process. Using a model that provides strong performance without unnecessary computational complexity can reduce processing time, energy consumption and infrastructure requirements. Retraining should therefore be performed when justified by meaningful data or performance changes rather than unnecessarily frequently. In a larger mobility system, the resource cost of continuous sensor data collection, cloud storage, model training and real-time inference should also be considered alongside the operational benefits.